

life expectancy

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Abstract

This study investigates whether life expectancy can be predicted using socio-economic indicators from Our World in Data. Several predictive models were developed, including linear regression, K-nearest neighbours, and random forests. Model performance was evaluated using RMSE. The random forest model achieved the best performance with a final test RMSE of 1.8 years. Fertility rate and healthcare expenditure emerged as the most influential predictors. GDP per capita was strongly correlated with life expectancy but added limited predictive value once other socio-economic indicators were included.

Introduction

The course HarvardX PH125.9x: Data Science Capstone is the ninth and final component of the Data Science Professional Certificate program. Within this course, students are required to complete two projects that together determine the final grade. This report describes the second project, in which students are free to select their own research topic. For this study, the chosen topic is an analysis of socio-economic factors that may influence life expectancy at birth.

The central research questions are:

1. To what extent can life expectancy at birth be predicted using socio-economic indicators?
2. Which socio-economic factors contribute most strongly to these predictions?
3. Which modelling approach provides the best predictive performance?

Life expectancy at birth represents the average number of years a newborn is expected to live. Over the past decades, substantial improvements in public health have been achieved worldwide. According to the World Health Organization (WHO), global life expectancy increased significantly between 2000 and 2019. However, this overall progress masks considerable differences between countries and population groups. Individuals with a lower socio-economic status often have less access to healthcare and education, which may negatively affect their health outcomes and life expectancy (World Health Organization, n.d.). In addition to access to healthcare and education, several other factors may contribute to differences in life expectancy. This study utilizes data obtained from Our World in Data (OWID) (Our World in Data, n.d.). The dataset contains a wide range of indicators related to economic conditions, health, lifestyle, and demographics.

The following variables are included in the analysis:

- Healthcare spending (% of GDP)
- Education spending (% of GDP)
- Years of schooling (average years of education)
- Vaccination coverage (multiple vaccination indicators)
- Obesity prevalence (share of adults classified as obese)

- Fertility rate (children per woman)
- Urbanization (urban and rural population shares)
- Renewable energy (share of total energy consumption from renewable sources)
- Democracy index (ranging from 0 to 1, where 1 represents the highest level of democracy)
- GDP per capita (in U.S. dollars)

The target variable in this study is life expectancy at birth. Using various data analysis techniques and predictive models, the study investigates which factors are most strongly associated with life expectancy and to what extent life expectancy can be predicted accurately.

Approach

The final dataset was constructed by combining eleven separate datasets obtained from Our World in Data (OWID). From each dataset, only the variables relevant to this study were selected. Subsequently, all datasets were transformed into a common structure, allowing them to be merged into a single integrated dataset. The healthcare dataset contained the smallest number of observations and was therefore used as the foundation for the merging process. This approach was chosen to minimize the number of missing values introduced during data integration. Despite this strategy, the combined dataset still contained a substantial amount of missing data.

Table 1: Variables with missing values

	Variable	Missing_Percentage
coverage__antigen_rotac	coverage__antigen_rotac	85.4
coverage__antigen_ipv1	coverage__antigen_ipv1	77.8
coverage__antigen_pcv3	coverage__antigen_pcv3	74.2
coverage__antigen_hepb3	coverage__antigen_hepb3	51.8
coverage__antigen_hib3	coverage__antigen_hib3	47.1
spending_edu	spending_edu	38.2
years_in_school	years_in_school	35.1
coverage__antigen_rcv1	coverage__antigen_rcv1	32.2
coverage__antigen_mcv1	coverage__antigen_mcv1	26.2
coverage__antigen_dtpcv3	coverage__antigen_dtpcv3	25.4
coverage__antigen_pol3	coverage__antigen_pol3	25.3
share_obesity	share_obesity	24.1
renewables_share_pct	renewables_share_pct	16.5
move_from_city	move_from_city	16.3
move_to_city	move_to_city	16.3
gdp_pc	gdp_pc	9.4
children_per_woman	children_per_woman	9.1
life_expectancy	life_expectancy	5.8
elect_dem_index	elect_dem_index	1.3

Variables with more than 50% missing values were removed from the analysis. The remaining missing values were imputed using the missForest algorithm after the data had been divided into training, validation, and final test sets.

This approach was selected because previous research has shown that missForest can effectively handle different data types and often outperforms several traditional imputation methods. An additional advantage is that the method provides an out-of-bag estimate of the imputation error (Stekhoven and Bühlmann, 2012). Furthermore, the relationships among variables in this dataset are likely to be both linear and non-linear,

making simple median or mean imputation less appropriate, as these methods may fail to preserve the underlying structure of the data.

An 85% / 15% split was used to obtain a sufficiently large independent final test set, while keeping enough observations for model training. The remaining 85% of the data was further divided into 80% training and 20% validation data. This resulted in approximately 68% of the original data being used for training, 17% for validation, and 15% for final testing.

To avoid data leakage, the imputation model was fitted exclusively on the training data and then applied to the validation and final test sets.

Following imputation and data preprocessing, no missing values remained in the training, validation, or final test sets.

No missing values were found in the training, validation, or final test set.

To assess whether the data partitions are representative of the original dataset, a boxplot was constructed. It shows that the distributions of life expectancy are similar across the training, validation, and final test sets, suggesting that the data split was successful.

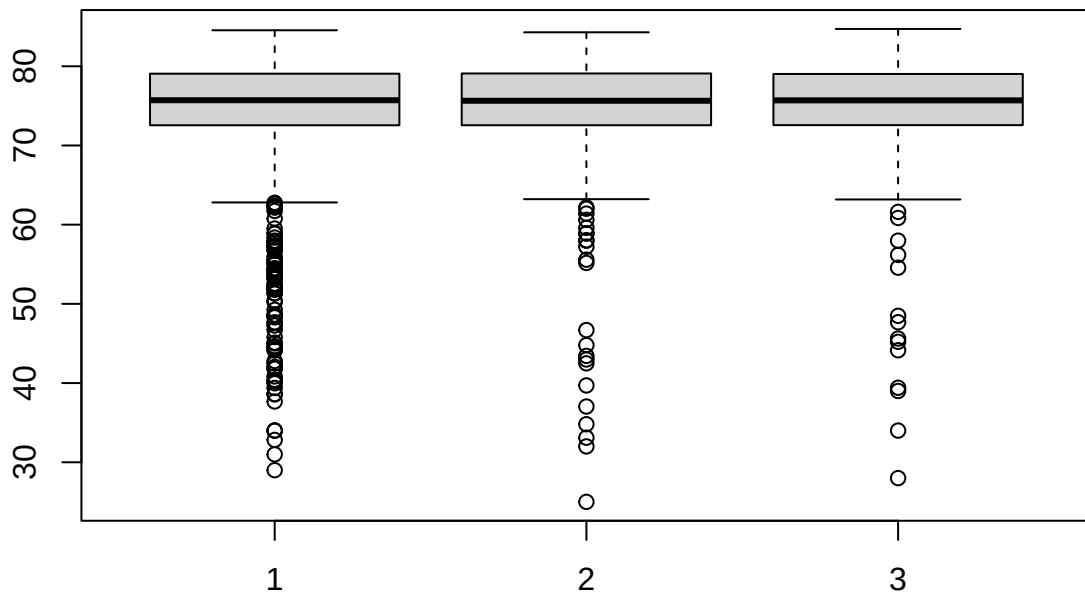


Table 2: Descriptive statistics of life expectancy across the data splits

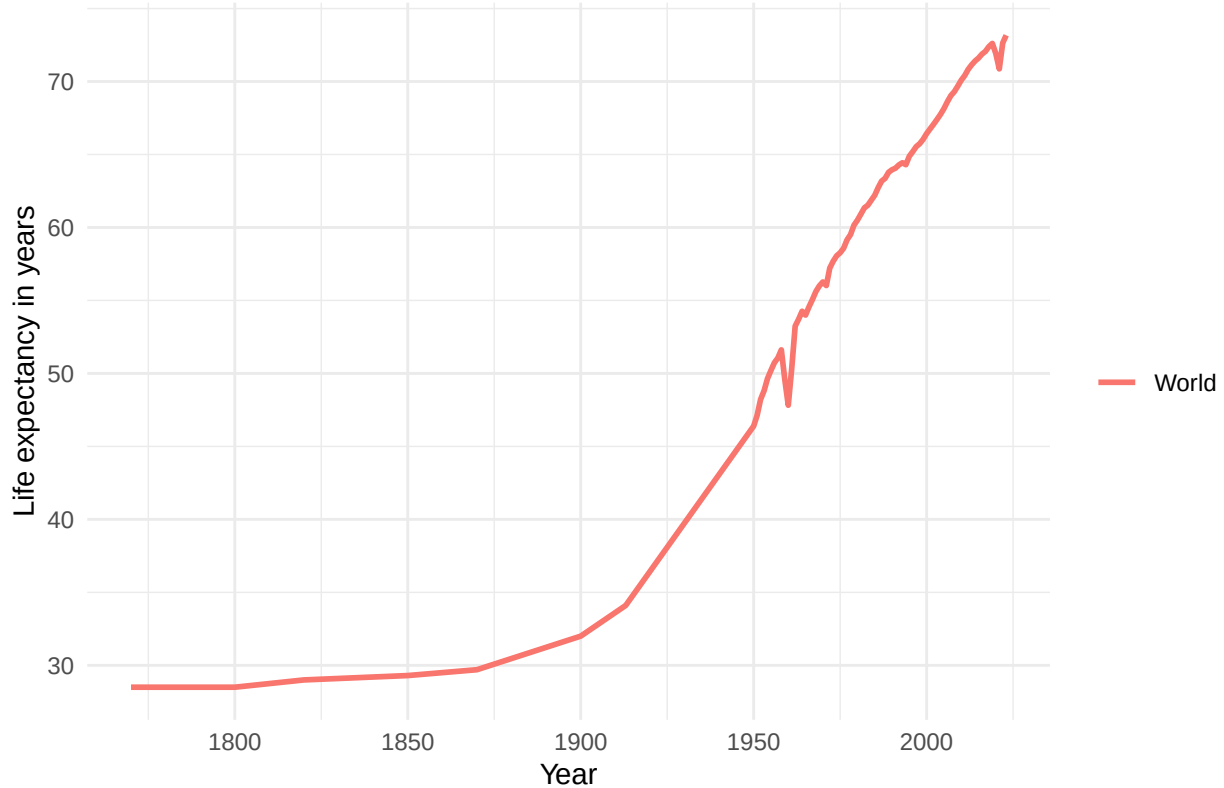
Dataset	N	Mean	SD	Min	Median	Max
Train	1632	74.62	7.35	29	75.71	84.55
Validation	407	74.47	7.88	25	75.65	84.29
Test	356	74.77	7.27	28	75.70	84.71

Dataset	N	Mean	SD	Min	Median	Max
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EDA

If we examine global life expectancy over time (Figure 1), it is evident that life expectancy has increased substantially over the past decades. Overall, this suggests that global health outcomes have improved considerably.

Fig 1. Life expectancy over time

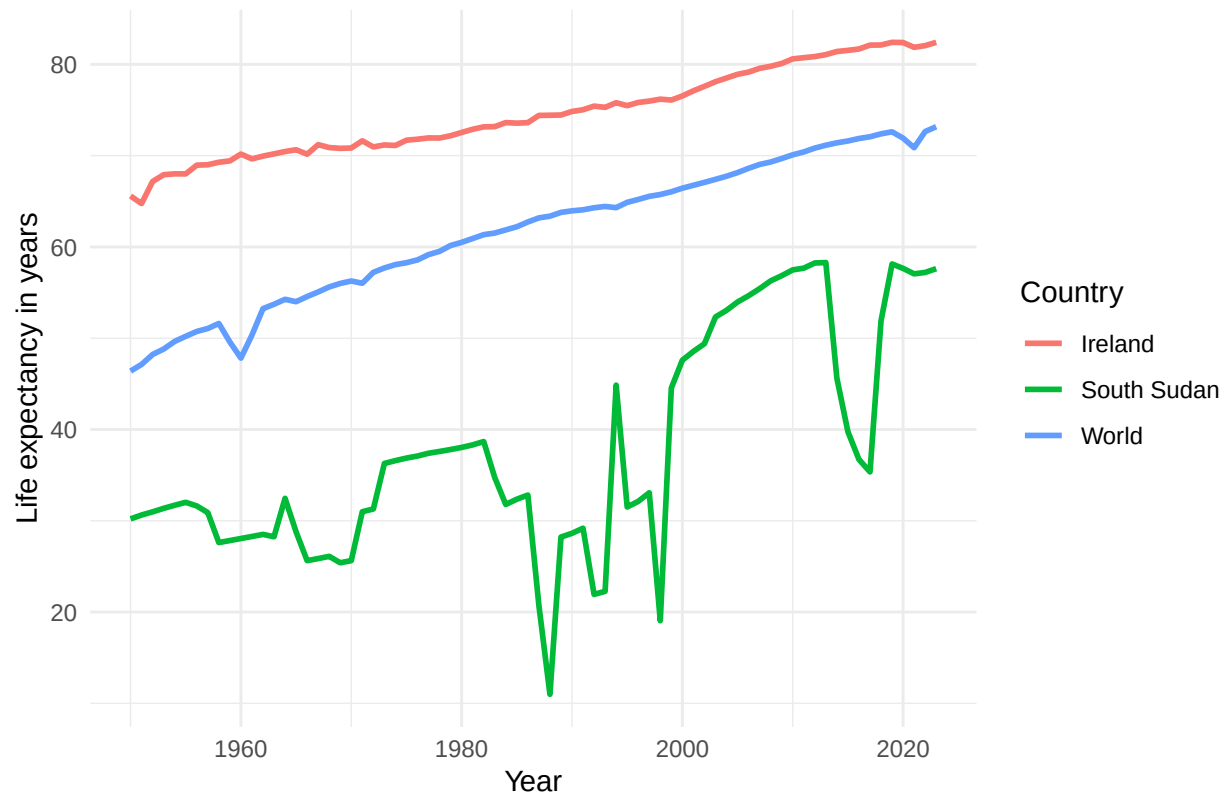


However, when comparing countries with the highest and lowest GDP per capita (Figure 2), a clear disparity becomes visible. The poorest country in the dataset has a life expectancy below 60 years, whereas the richest country has a life expectancy exceeding 80 years. This represents a difference of more than 20 years. Furthermore, when compared with the global average life expectancy of over 70 years, the poorest country falls approximately 10 years below the worldwide average.

Table 3: Countries with the Highest and Lowest GDP per Capita

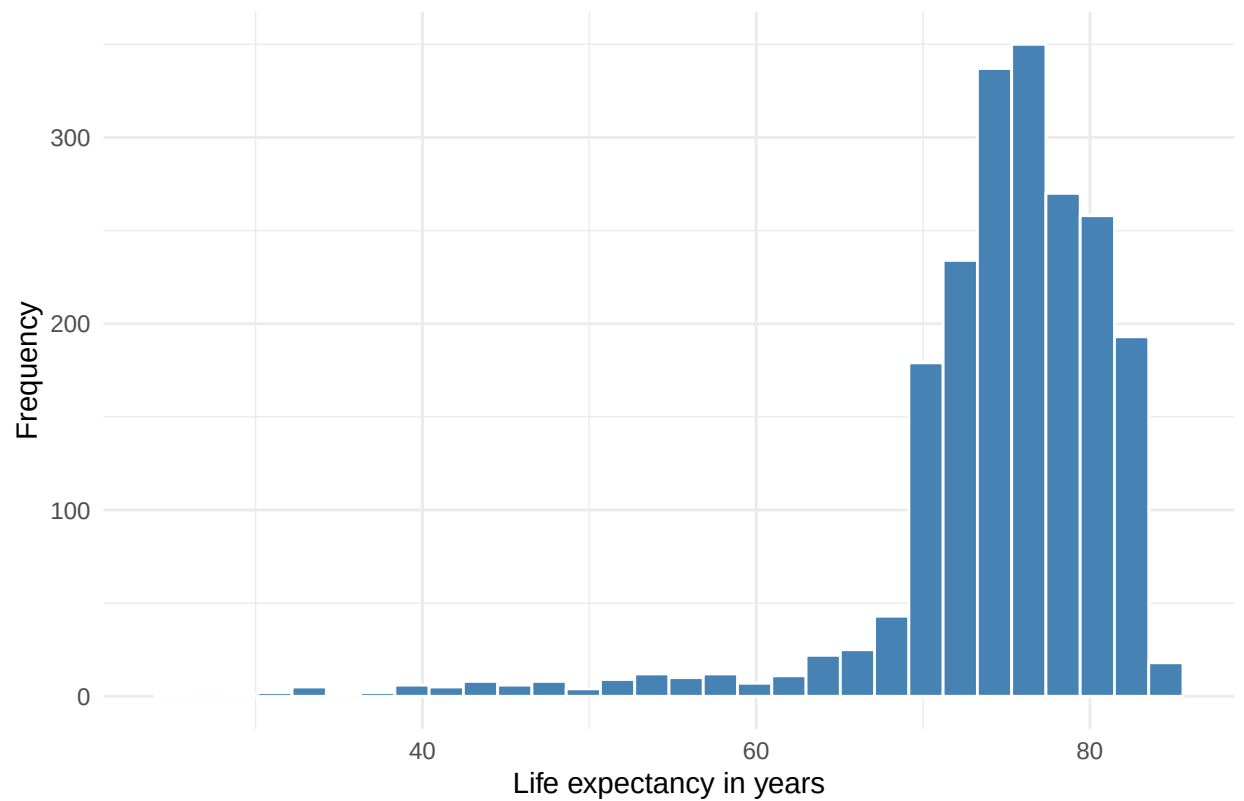
Category	Country	Year	GDP per Capita
Richest Country	Ireland	2023	132543.2300
Poorest Country	South Sudan	2023	727.0306

Fig 2. life expectancy poor versus rich



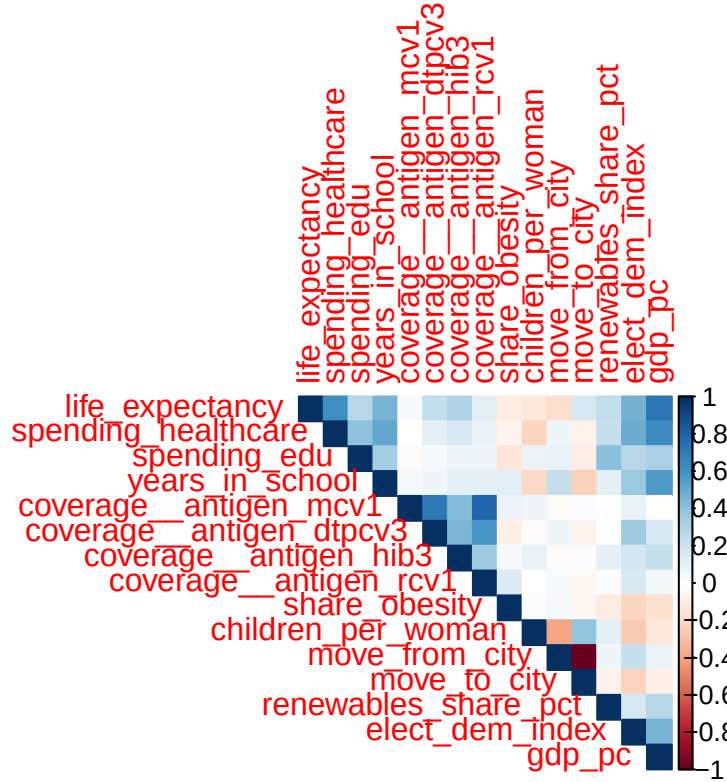
If we look at the target variable (life expectancy) (Figure 3), it becomes clear that the distribution is right-skewed. This indicates substantial variation across observations and suggests that many countries cluster around relatively high life expectancy values while a smaller number of countries have considerably lower values.

Fig 3. Distribution of target variable



A correlation matrix (Figure 4) reveals that the target variable is strongly correlated with several predictors, most notably GDP per capita, suggesting that economic conditions may be an important predictor of life expectancy and that GDP per capita itself contains substantial information about the target variable.

fig 4. Correlation predictors

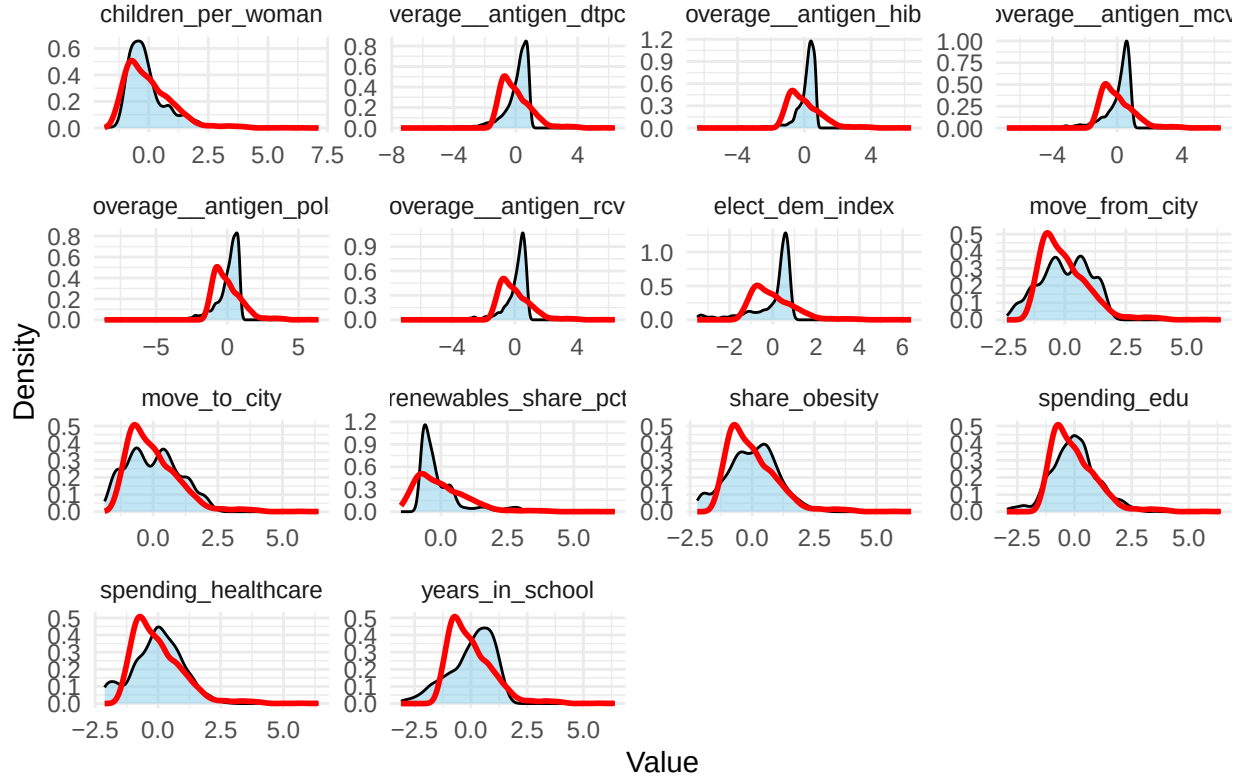


However, GDP per capita is also strongly correlated with several other predictors, suggesting that it may capture broader socio-economic conditions reflected in multiple indicators. Figure 5 shows that, after standardization, the distribution of GDP per capita overlaps substantially with the distributions of many predictors. While this overlap does not imply redundancy on its own, it suggests that GDP per capita may share information with other variables in the dataset.

For instance, healthcare spending is strongly correlated with GDP per capita. If we look at the distribution of healthcare spending and GDP per capita, it is noticeable that GDP covers a big area of healthcare spending, which could indicate that some predictive information may already be captured by other predictors in the dataset.

Therefore, we hypothesize that the incremental predictive value of GDP per capita may be smaller than expected once the remaining predictors are included in the model. This hypothesis will be evaluated in the modelling section by comparing model performance and variable importance measures.

fig 5. Distribution of predictors with GDP per capita as overlay



Modelling

Before building the predictive models, it is important to consider which variables should be included in the analysis. The objective of this study is to identify socio-economic factors that are associated with life expectancy at birth.

Variables such as the country itself contribute little to understanding these underlying relationships. If a model learns that a particular country tends to have a higher or lower life expectancy, country-specific characteristics may obscure the effects of the actual predictors of interest. Therefore, the country variable was removed from the dataset prior to model training.

In addition, the year variable was excluded from the final models. The purpose of this study is to identify socio-economic factors associated with life expectancy, rather than to capture temporal trends. The year itself is not a direct explanatory factor but rather reflects broader societal, medical, and technological developments that have occurred over time. Excluding year allows the interpretation of the remaining predictors to remain more closely aligned with the research objective. Furthermore, it prevents the models from deriving a substantial portion of their predictive performance from the general upward trend in life expectancy over time, rather than from the socio-economic factors under investigation.

The modelling process begins with a simple baseline model, followed by a linear regression model, a k-nearest neighbours (KNN) model, and a Random Forest model. Model performance is evaluated using the Root Mean Squared Error (RMSE) metric.

The RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y} - y)^2}$$

where:

- y Actual life expectancy
- $\hat{y}$ Predicted life expectancy
- n Number of observations

The RMSE measures the average magnitude of the prediction errors, giving greater weight to large errors due to the squaring operation. A lower RMSE indicates a more accurate predictive model.

To assess the predictive contribution of GDP per capita, the modelling procedure is conducted twice: once excluding GDP per capita and once including it as a predictor. As suggested in the exploratory data analysis, GDP per capita may share much of its explanatory power with other socio-economic variables. Comparing model performance under both scenarios will therefore provide insight into the unique predictive value of GDP per capita.

Finally, the best-performing model, as determined by the validation RMSE, will be evaluated on the independent final test set to obtain an unbiased estimate of its predictive performance.

A simple baseline model yields an RMSE of approximately 7.9 years. This result is based solely on using the mean life expectancy as the prediction.

```
## [1] 7.866865
```

The linear model without GDP per capita already performs considerably better, reducing the RMSE to 4.5 years. Figure 6 shows the relationship between actual and predicted life expectancy. An interesting question is which variables contribute most to the model's performance. Fertility, measured as the number of children per woman, is by far the most important predictor.

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## [1] 4.522634
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Fig 6.: Actual vs predicted (linear)

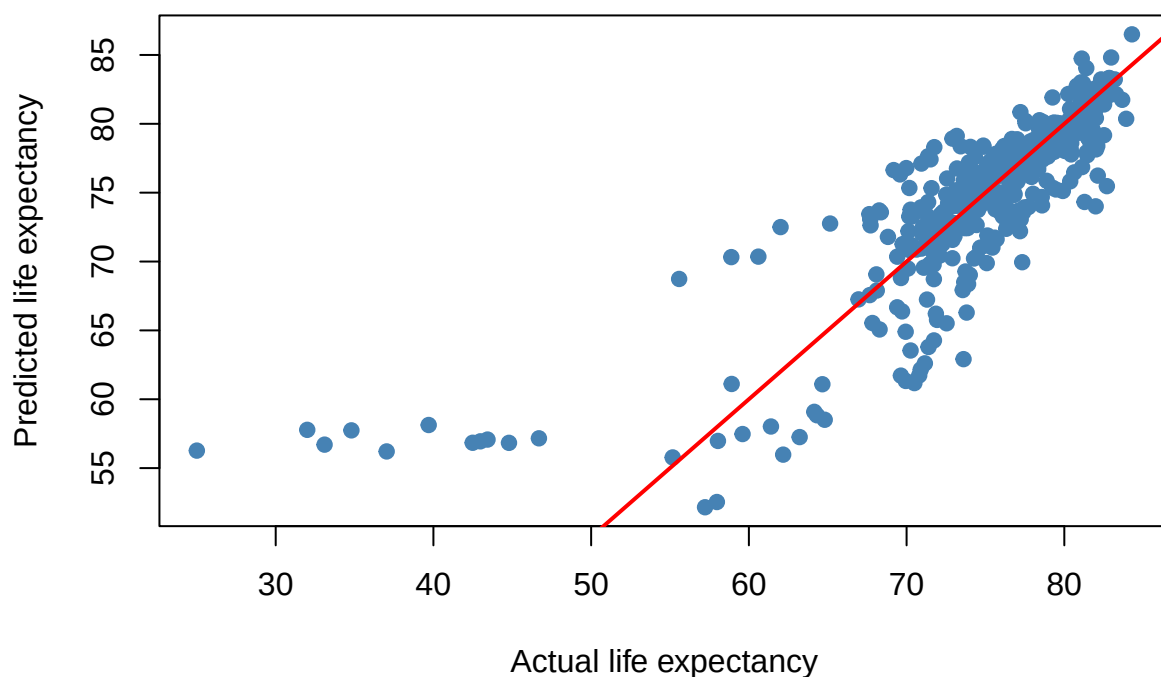


Table 4: Regression Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	35.282	63.188	0.558	0.577
spending_healthcare	0.959	0.077	12.426	0.000
spending_edu	-0.300	0.109	-2.752	0.006
years_in_school	0.068	0.072	0.936	0.349
coverage__antigen_mcv1	0.031	0.025	1.245	0.213
coverage__antigen_dtpcv3	0.055	0.016	3.541	0.000
coverage__antigen_hib3	0.075	0.009	8.040	0.000
coverage__antigen_rcv1	-0.014	0.024	-0.572	0.567
share_obesity	-0.050	0.016	-3.228	0.001
children_per_woman	-3.266	0.198	-16.522	0.000
move_from_city	0.186	0.632	0.294	0.769
move_to_city	0.289	0.632	0.458	0.647
renewables_share_pct	0.051	0.010	5.187	0.000
elect_dem_index	5.813	0.656	8.857	0.000

When the same model is estimated with GDP per capita included, the RMSE changes very little. Fertility remains the most important variable, indicating that GDP per capita adds limited predictive value beyond the information already captured by the other variables.

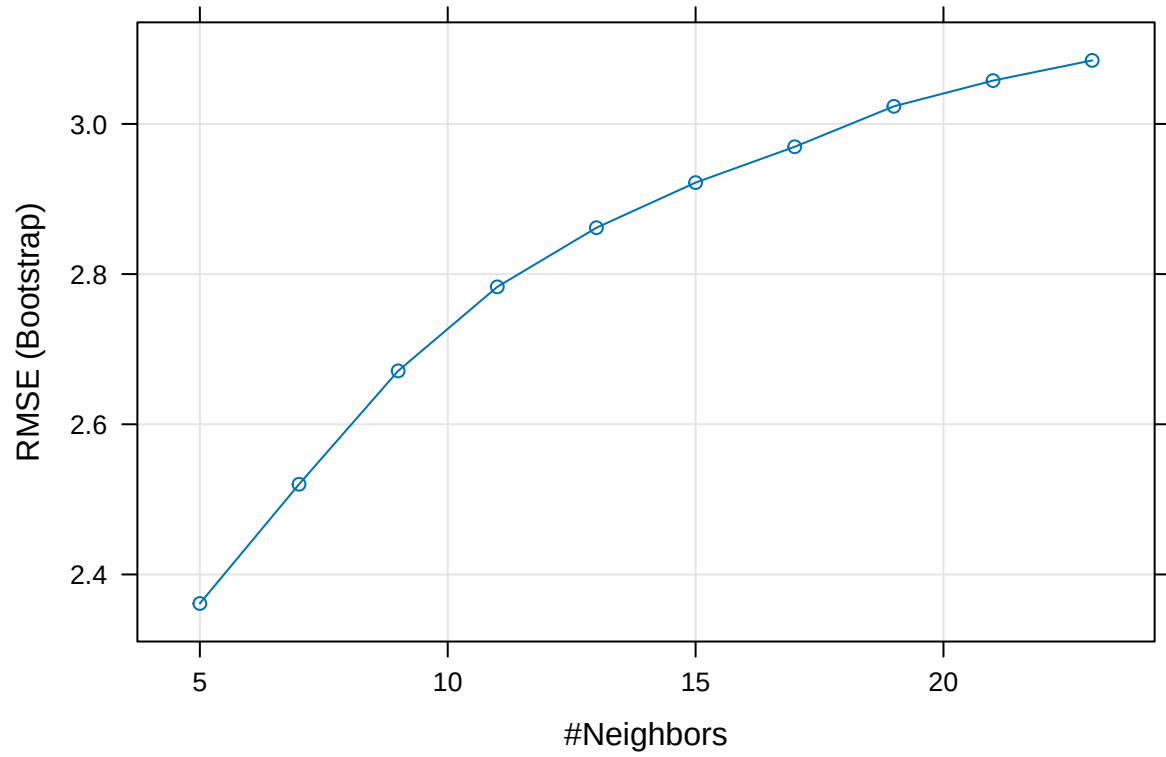
[1] 4.494382

Table 5: Regression Model Results with GDP

term	estimate	std.error	statistic	p.value
(Intercept)	99.182	62.954	1.575	0.115
spending_healthcare	0.755	0.082	9.252	0.000
spending_edu	-0.202	0.108	-1.860	0.063
years_in_school	-0.152	0.078	-1.952	0.051
coverage__antigen_mcv1	0.027	0.024	1.105	0.269
coverage__antigen_dtpcv3	0.050	0.015	3.236	0.001
coverage__antigen_hib3	0.069	0.009	7.495	0.000
coverage__antigen_rcv1	-0.010	0.024	-0.402	0.687
share_obesity	-0.030	0.016	-1.939	0.053
children_per_woman	-3.419	0.196	-17.435	0.000
move_from_city	-0.423	0.629	-0.672	0.501
move_to_city	-0.328	0.629	-0.521	0.603
renewables_share_pct	0.043	0.010	4.438	0.000
elect_dem_index	5.100	0.655	7.786	0.000
gdp_pc	0.000	0.000	6.960	0.000

Using the same set of predictors in a K-nearest neighbors (KNN) model, but excluding GDP per capita, results in an RMSE of 2.3 years. Figure 7 illustrates the relationship between actual and predicted life expectancy. Compared with the linear model, the KNN model performs more accurately, suggesting that the relationship between the predictors and life expectancy is captured more effectively by a KNN model. In the KNN model, healthcare expenditure appears to be the most important predictor.

```
##      k
## 1  5
```



```
## [1] 2.270209
```

Figure 7: Actual vs predicted (KNN)

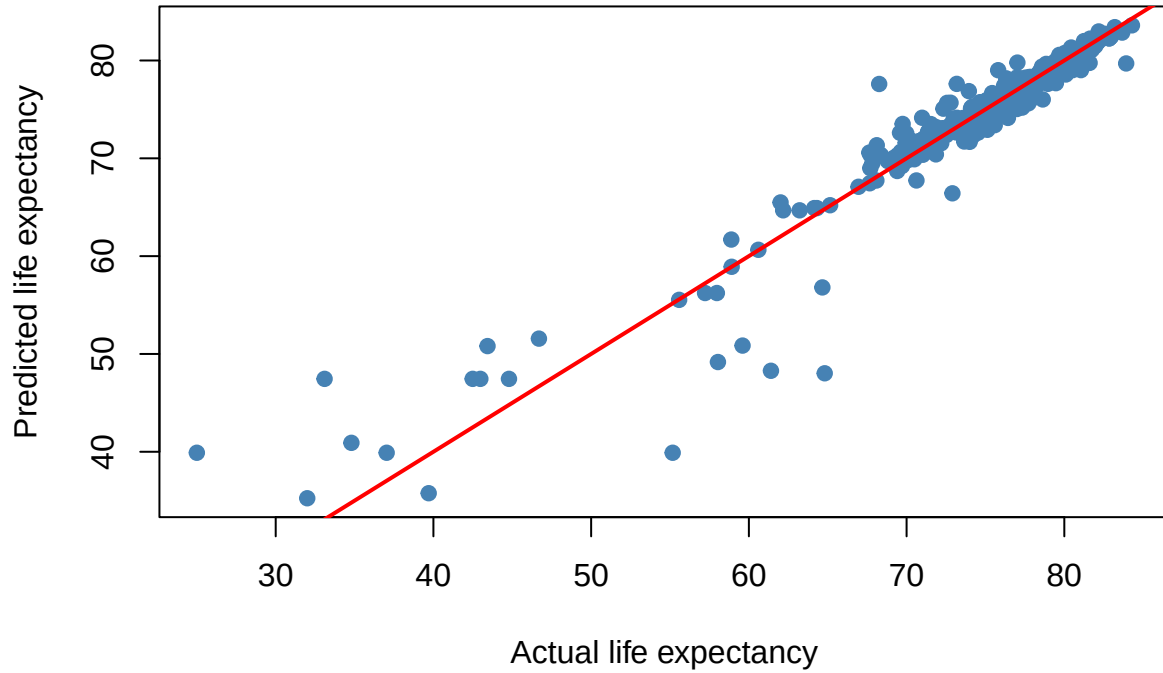
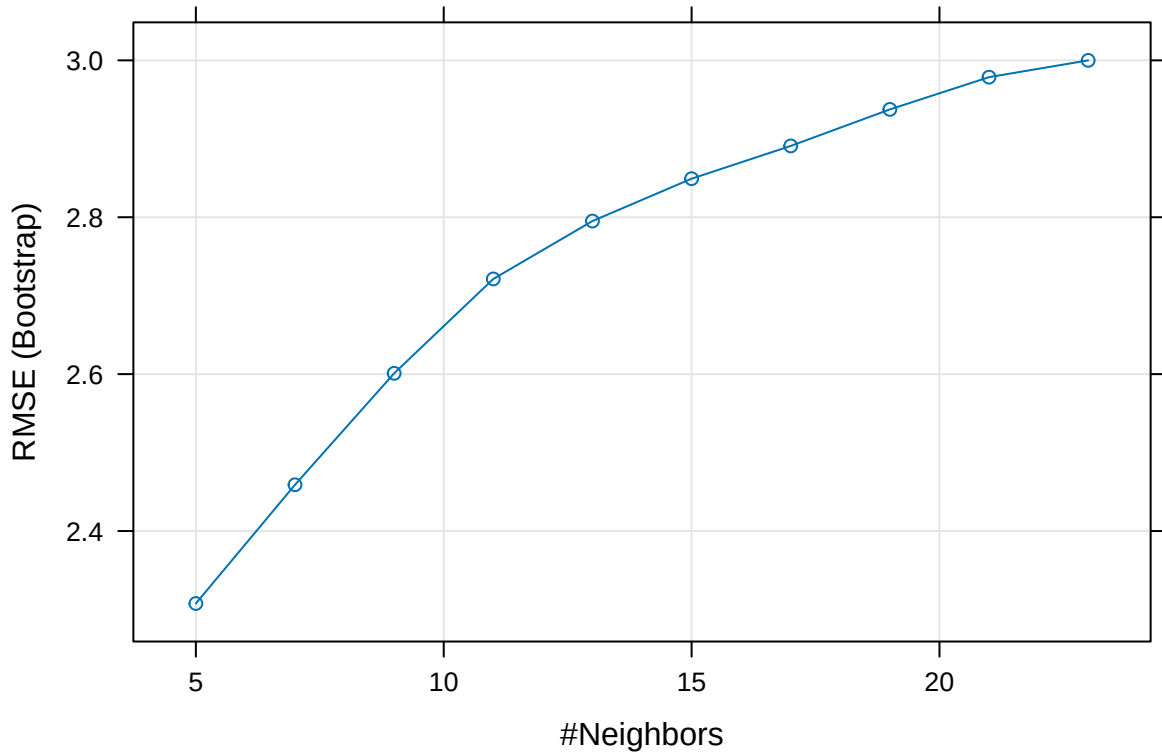


Table 6: Table 6. Variable importance for the KNN model

Variable	Overall
spending_healthcare	100.000
children_per_woman	67.661
coverage__antigen_dtpcv3	63.869
coverage__antigen_hib3	59.561
coverage__antigen_rcv1	57.049
coverage__antigen_mcv1	50.951
spending_edu	49.151
years_in_school	41.613
elect_dem_index	41.551
move_to_city	3.342
move_from_city	3.103
share_obesity	2.561
renewables_share_pct	0.000

When GDP per capita is added to the KNN model, the RMSE shows little to no improvement. However, GDP per capita exhibits greater predictive importance in this specification than in the linear model. Despite its increased contribution, the overall predictive performance of the model remains largely unchanged.

```
## k
## 1 5
```



[1] 2.214762

Table 7: Table 7. Variable importance for the KNN model with GDP

Variable	Overall
spending_healthcare	100.000
gdp_pc	77.631
children_per_woman	67.661
coverage__antigen_dtpcv3	63.869
coverage__antigen_hib3	59.561
coverage__antigen_rcv1	57.049
coverage__antigen_mcv1	50.951
spending_edu	49.151
years_in_school	41.613
elect_dem_index	41.551
move_to_city	3.342
move_from_city	3.103
share_obesity	2.561
renewables_share_pct	0.000

If we apply the same procedure using a Random Forest model, the resulting RMSE is 2 years. This performance is slightly better than that of the KNN model.

Figure 8 illustrates the relationship between actual and predicted life expectancy.

In the Random Forest model, healthcare expenditure and fertility rate emerge as the most important predictors.

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## [1] 2.00896
```

Figure 8: Actual vs predicted (RF)

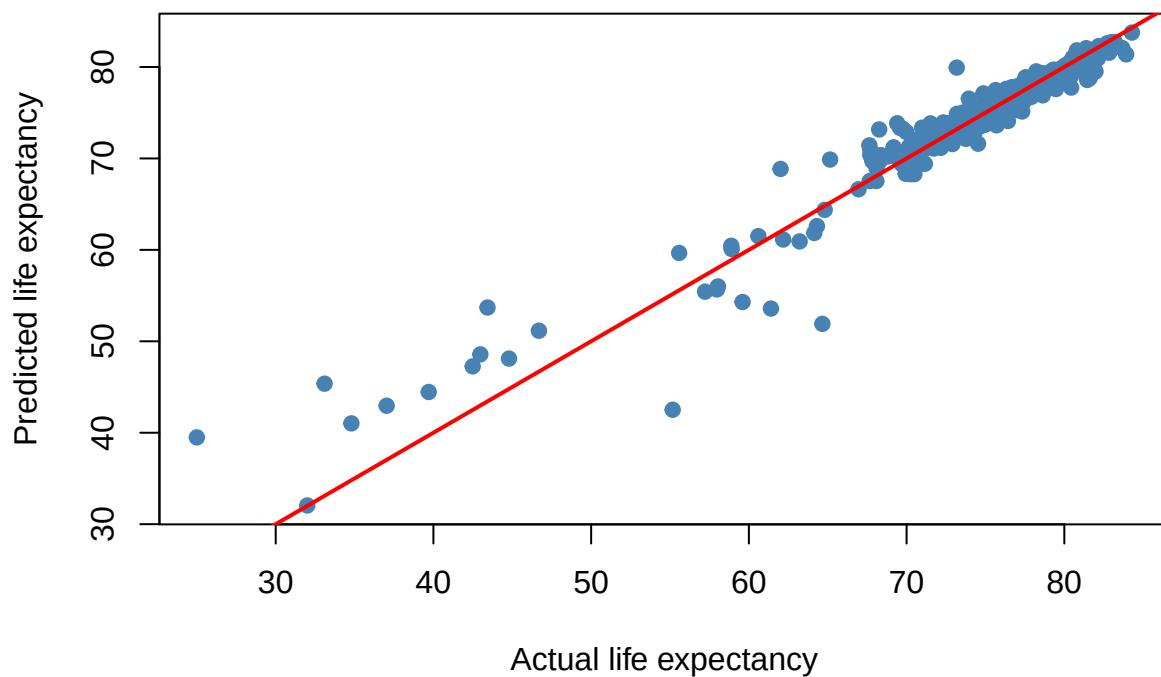
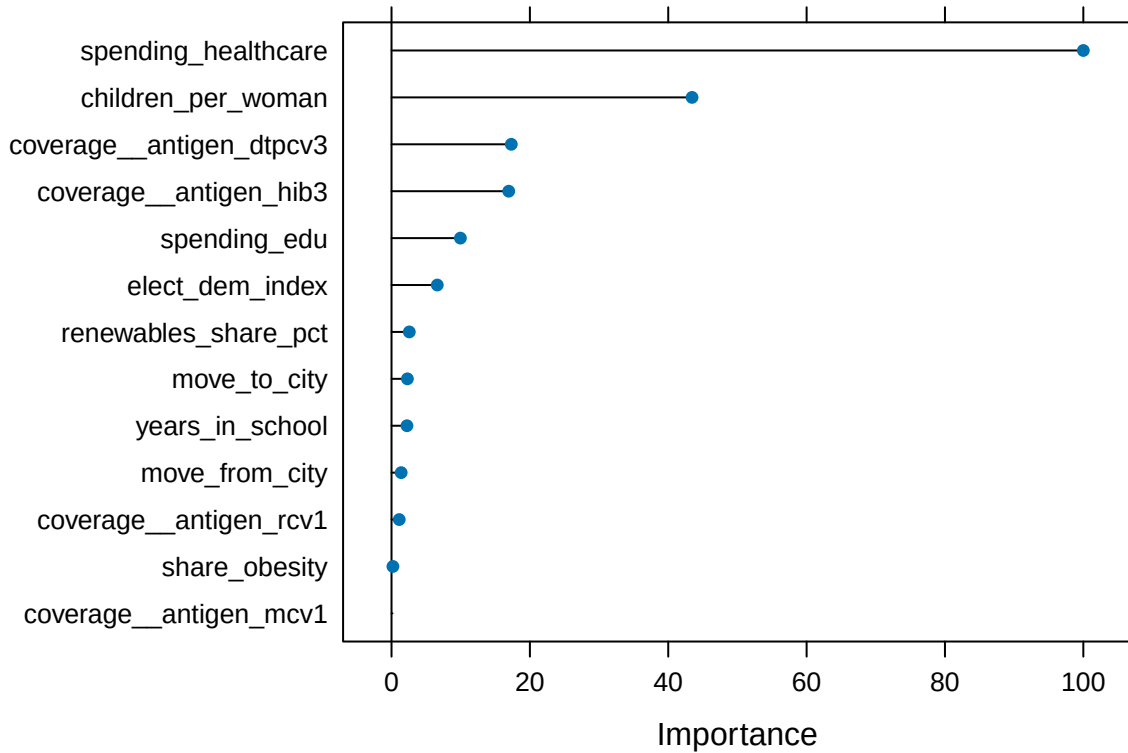


Table 8: Table 8. Variable importance for the RF model

Variable	Overall
spending_healthcare	100.000
children_per_woman	43.440
coverage__antigen_dtpcv3	17.316
coverage__antigen_hib3	16.944
spending_edu	9.955
elect_dem_index	6.615
renewables_share_pct	2.573
move_to_city	2.298
years_in_school	2.232
move_from_city	1.395
coverage__antigen_rcv1	1.113
share_obesity	0.202
coverage__antigen_mcv1	0.000



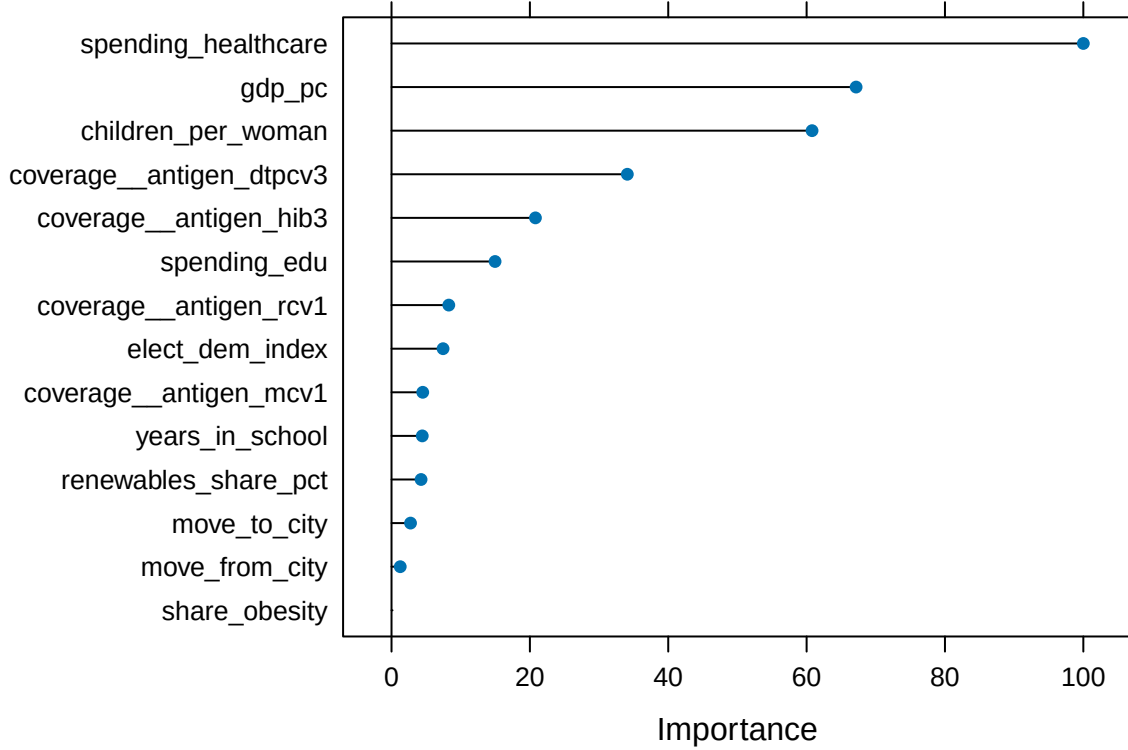
When GDP per capita is added to the Random Forest model, the RMSE does not improve that much. It settles around 1.98 years. Similar to the KNN results, GDP per capita appears to have a higher variable importance within the model, but its inclusion does not lead to a meaningful improvement in predictive performance. This finding suggests that much of the information contained in GDP per capita is already captured by the other socio-economic indicators included in the analysis.

```
## [1] 1.982481
```

Table 9: Table 9. Variable importance for the RF model with GDP

Variable	Overall
spending_healthcare	100.000
gdp_pc	67.157
children_per_woman	60.787
coverage__antigen_dtpcv3	34.086
coverage__antigen_hib3	20.805
spending_edu	14.969
coverage__antigen_rcv1	8.281
elect_dem_index	7.453
coverage__antigen_mcv1	4.513
years_in_school	4.444
renewables_share_pct	4.273
move_to_city	2.747
move_from_city	1.260
share_obesity	0.000

Variable	Overall
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Variable Importance. Based on the variable importance measures of the best performing model, the Random Forest model including GDP per capita, it became evident that the variable share of obesity contributed little to no predictive value. As its importance was effectively zero, this variable was excluded from the final model.

Although GDP per capita did not substantially improve predictive performance, the Random Forest model including GDP per capita was selected for evaluation on the independent final test set. This model achieved an RMSE of 1.8 years, indicating that life expectancy can be predicted with a relatively high degree of accuracy using the selected socio-economic indicators.

```
## [1] 1.827054
```

Table 9 summarizes the predictive performance of all models evaluated on the validation set. The results show a substantial improvement over the baseline model for all machine learning approaches. The Random Forest model achieved the lowest validation RMSE and was therefore selected for final evaluation on the independent test set. (Table 10)

Table 10: Table 10. Validation RMSE for all evaluated models

Model	Validation_RMSE
Baseline	7.87
Linear Regression	4.52

Model	Validation_RMSE
Linear Regression + GDP	4.49
KNN	2.27
KNN + GDP	2.21
Random Forest	2.01
Random Forest + GDP	1.98

Table 11: Table 11. Performance of the final selected model

Model	Validation_RMSE	Final_Test_RMSE
Random Forest + GDP	1.98	1.83

Although only a limited number of predictors are included in this analysis, a model such as this could still provide useful insights for policymakers seeking to improve population health outcomes and increase life expectancy. Additional health, environmental, and social indicators could also be incorporated to determine whether prediction accuracy can be improved further.

Conclusion

This study examined whether life expectancy at birth can be predicted using a range of socio-economic indicators and investigated which factors contribute most strongly to these predictions. Referring to the first and third research questions, the results show that life expectancy can be predicted with reasonable accuracy using socio-economic indicators. The best-performing model was a RF model, which achieved a final test RMSE of 1.8 years. This indicates that life expectancy can be predicted within approximately two years on average, representing a substantial improvement over the baseline model. With respect to the second research question, healthcare expenditure and fertility rate emerged as the most influential predictors across several modelling approaches. Education, vaccination cover-age, and other socio-economic indicators also contributed to the predictive performance of the models. An interesting finding is that GDP per capita, despite being strongly associated with life expectancy, added relatively little predictive value once other socio-economic indicators were included in the analysis. This suggests that the effects of national wealth may already be reflected through factors such as healthcare, education, and other investments that directly influence population health.

Several limitations should be acknowledged. The analysis was restricted to the variables available from the selected data sources, and substantial missing data required imputation. Furthermore, the results represent statistical associations and should not be interpreted as causal relationships. Overall, the findings suggest that socio-economic indicators can provide meaningful insights into differences in life expectancy. While economic prosperity remains important, the way resources are translated into healthcare, education, and broader social development appears to be at least as important as income levels alone.

References

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Use of Artificial Intelligence

Microsoft Copilot was used to assist with spelling and grammar checks and to provide suggestions for structuring the project and identifying potential analysis approaches. All data collection, coding, modelling, interpretation of results, and final writing decisions were performed by the author.